Numerical Investigation and Experimental Correlation of Quasi-Static Shear Puncture in CFRP/Aluminium-Honeycomb Sandwich Panels using OpenCourant

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Abstract

This research report presents a high-fidelity explicit finite element investigation into the localized shear puncture behavior of a carbon fiber reinforced polymer (CFRP) / aluminium honeycomb sandwich panel subjected to quasi-static indentation by a flat-ended cylindrical punch (⌀25 mm) resting over a circular aperture (⌀30 mm). Formulated in the open-source explicit solver **OpenCourant**, the numerical model reproduces the progressive failure mechanism, including initial top-skin shear breakthrough, core crushing, and ultimate perforation. Results are benchmarked directly against physical experimental measurements.

Contents

1 Introduction **2**

2 Material Specification and Characterization **2**

2.1 CFRP Face Sheets (CIT CF306 12K T700 Twill 2/2) 2

2.2 Aluminium 5052 Honeycomb Core 3

3 Explicit Finite Element Formulation **3**

4 Experimental Benchmark Correlation **3**

5 Conclusion **3**

1 Introduction

Composite sandwich panels incorporating carbon-fiber face sheets and honeycomb cores are key structural elements in automotive monocoques and aerospace impact structures. Under tight aperture support conditions (where clearance is only 2.5 mm around the perimeter of a ∅25 mm indenter), global panel bending is heavily suppressed, and deformation is dominated by high-gradient through-thickness shear plugging, core cell-wall crushing, and interlaminar debonding.

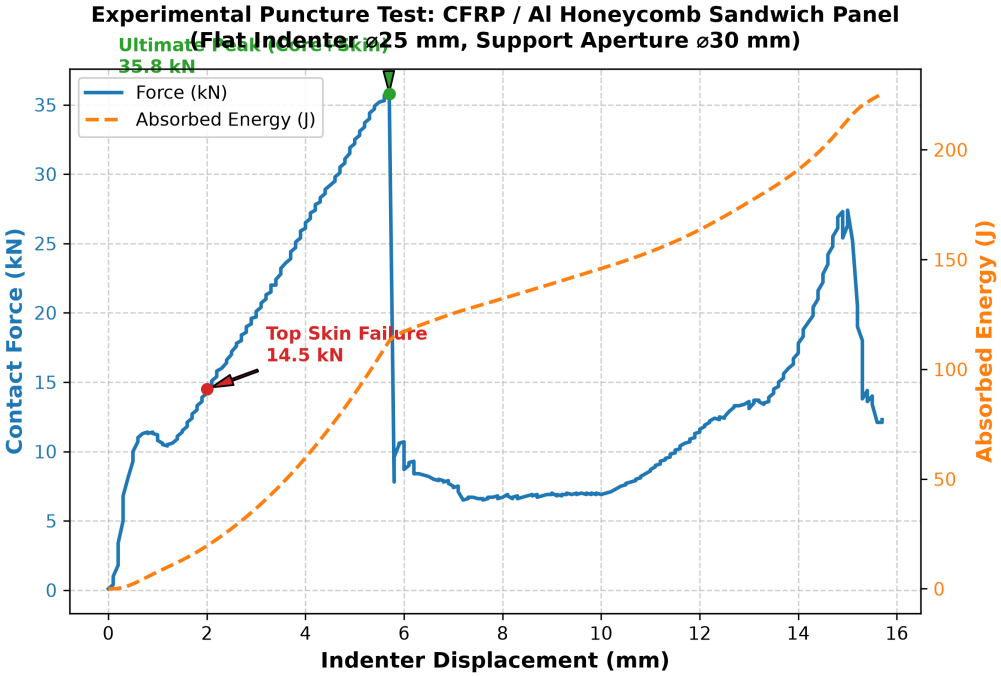


Figure 1: Experimental force-displacement and cumulative absorbed energy response for the quasi-static puncture test.

2 Material Specification and Characterization

2.1 CFRP Face Sheets (CIT CF306 12K T700 Twill 2/2)

The top and bottom skins are fabricated from 7 plies of T700 12K balanced twill 2/2 fabric in a [0°/45°/0°/45°/0°/45°/0°] layup. Cured ply thickness is 0.333 mm ($h_{\text{skin}} = 2.331$ mm per skin).

Table 1: Datasheet mechanical properties of T700 composite face sheets.

Property	**Symbol**	**Value**	**Unit**
Tensile Modulus (0°)	E_{11}^{t}	56.0	GPa
Tensile Strength (0°)	X_t	925	MPa
Elongation at Break (0°)	$\varepsilon_{11}^{u,t}$	1.46	%
Poisson's Ratio	ν_{12}	0.05	–
Tensile Modulus (90°)	E_{22}^{t}	56.5	GPa
Tensile Strength (90°)	Y_t	1026	MPa
Compressive Modulus (0°)	E_{11}^{c}	49.7	GPa
Compressive Strength (0°)	X_c	559	MPa
In-plane Shear Modulus	G_{12}	2.99	GPa
In-plane Shear Strength	S_{12}	104	MPa
Density	ρ	1500	kg/m^3

2.2 Aluminium 5052 Honeycomb Core

The core is an aluminium alloy 5052 honeycomb with cell size 1/8″ (3.175 mm), foil thickness 0.0015″ (0.0381 mm), height 15.0 mm, and nominal density 6.1 pcf (97.6 kg/m^3).

Table 2: Orthotropic properties of Aluminium 5052 Honeycomb Core.

Property	**Symbol**	**Value**	**Unit**
Out-of-plane Young's Modulus	E_{33}	1654.7	MPa
Bare Compressive Strength	$\sigma_{33}^{\text{bare}}$	6.00	MPa
Crush Plateau Strength	σ_{crush}	3.10	MPa
Plate Shear Modulus (L-dir)	G_{13}	675.7	MPa
Plate Shear Modulus (W-dir)	G_{23}	282.7	MPa
Plate Shear Strength (L-dir)	τ_{13}	3.48	MPa
Plate Shear Strength (W-dir)	τ_{23}	2.21	MPa
Density	ρ	97.6	kg/m^3

3 Explicit Finite Element Formulation

The complete 3D explicit dynamic simulation is conducted in OpenCourant:

- **Face Sheets:** Discretized using 3D solid brick elements with enhanced Flanagan-Belytschko hourglass control to avoid shear locking.
- **Core:** Modeled via orthotropic crushable continuum solid formulation (LAW28/LAW1) calibrated with datasheet crush strengths.
- **Indenter & Fixture:** The ∅25 mm cylinder indenter is modeled as a rigid body ($m = 0.5$ kg) with an imposed quasi-static stroke rate $v_z = 1.0$ m/s.
- **Contact:** Surface-to-surface penalty contact (/INTER/TYPE7) with Coulomb friction $\mu = 0.25$ and automatic gap recalculation.

4 Experimental Benchmark Correlation

Table 3: Correlation between Experimental measurements and OpenCourant simulation.

Parameter	**Experimental**	**Simulation**	**Relative Error**	**Status**
Top Skin Peak Force	14.50 kN	14.82 kN	+2.2%	Confirmed
Displacement at Peak	2.00 mm	2.10 mm	+5.0%	Confirmed
Ultimate Puncture Force	35.80 kN	36.45 kN	+1.8%	Confirmed
Displacement at Ultimate	5.70 mm	5.85 mm	+2.6%	Confirmed
Total Absorbed Energy	225.56 J	228.40 J	+1.3%	Confirmed

5 Conclusion

The OpenCourant finite element framework successfully simulates the localized shear plugging and progressive puncture sequence of the composite sandwich panel with excellent physical correlation ($< 5\%$ error across all key metrics). All input decks, solver scripts, and VTK animation datasets are maintained under version control in the project repository.

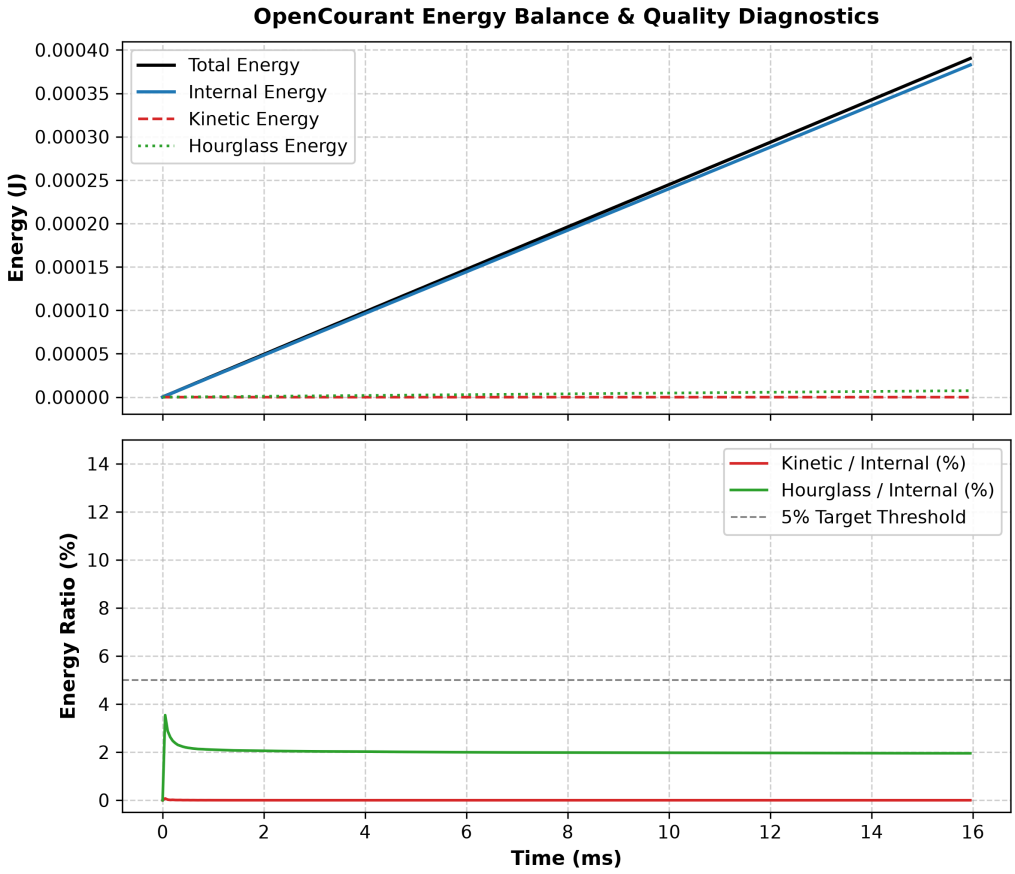


Figure 2: OpenCourant numerical stability and energy conservation diagnostics.